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DDT File

Sanity in Research and Evaluation of Environmental Health

How to achieve a realistic evaluation (in seven commandments).

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Early in 1969, before the methyl mercury story broke into headlines, I presented a Cummings Memorial Lecture titled, "The Spectre of Today's Environmental Pollution—U.S.A. Brand—New Perspectives from an Old Scout" (1). I attempted a sane and realistic, comprehensive toxicologic evaluation of the potential hazards to human health from man-made and natural environmental pollutants (except radiation) in air, water, and food, based on their occurrence in the continental United States then and in the foreseeable future. The evaluation was made on a "before and after" consideration of all those substances about which health-oriented groups had expressed concern. After a considered appraisal of the available facts, I judged the validity of the concern for each individual pollutant.

Of 18 major pollutants or groups of pollutants from all sources (air, water, and food) combined, only eight were reevaluated as potentially hazardous, and not all of these with the same degree of certainty. There was no question, for example, of the deleterious effects of environmental nitrates (and nitrites) on infants and on genetically hypersusceptible Alaskan Eskimos and Indians, but the hazards to public health of "hard" water constituents, while suggestive, were far from proven. Respiratory irritants (particularly oxidants) and carcinogens from all three

environmental sources, either alone or in combination with respiratory irritants, headed the list as pollutants of primary concern; yet pesticides of all types, as well as teratogens and mutagens, as found in the three environmental sources, were conspicuously absent from the list. Asthmagens were not absent.

What has the record shown since? The pesticide DDT [1,1,1-trichloro-2,2-bis(p-chlorophenyl)ethane] and the herbicide 2,4,5-T (2,4,5-trichlorophenoxyacetic acid) were, on first action, banned from further use, as were all uses of organic mercurials as pesticides. Swordfish has been removed from sale, although people may still eat fish, just as they have since before the time of Christ (2). Cyclamates have gone, and monosodium glutamate is on its way out. Nitrilotriacetic acid (NTA), the soap industry's dream substitute for the polyphosphates that the industry itself banned from laundry detergents, was summarily prohibited on fragile and provocative evidence (3). Detergent enzymes appear to be on their way out. Potato chips may be next because of their increased solanin content, caused by dehydration of hot fat.

What next? If man is prohibited from consuming foodstuffs because of their content of natural toxins, where will it end? Honey contains the potent grayanotoxins (given intraperitoneally, a dose lethal to 50 percent of the animals tested is approximately 1 milligram per kilogram) (4). German raisins have been found to be teratogenic (5), while caffeine (6) and tannin are tumorigenic (7); nutmeg, parsley, and dill are highly toxic because of their myristicin and apiol contents (8). Vitamin C has produced tumors in mice (9).

The following rules, if followed, should go far in improving the criteria and standards for judging environmental pollution, as well as making them less worrisome to those who are being protected and more acceptable to those who must comply.

Commandment 1

Standards must be based on scientific facts, realistically derived, and not on political feasibility, expedience, emotion of the moment, or unsupported information. If necessary data are not available, studies should be made to supply them. Meanwhile, provisional, tentative, or best judgment standards, clearly marked and recognized as such, should be proposed, but only upon definite need. If the need is not there, it is better to withhold until such time as the facts are in.

This commandment poses relatively few difficulties for drinking water standards. Based on the accepted average daily intake of 2 liters per person (in the United States), and primarily on human health or esthetic considerations, standards have been and can be determined reasonably well. For toxic food contaminants, the problems are more numerous but not unresolvable. In this case, because of widely varied eating habits, realistic evaluation of the facts (generally derived from animals) requires more erring on the side of common sense than on the side of the over-protection complex that seems to have captivated the thinking of those committed to setting tolerances. In the case of air pollutants, scientific facts for each of the multiple criteria (for example, human health, nuisance, crop and vegetation damage, surface erosion, and soiling) must be carefully developed and painstakingly assessed before air quality standards are proposed and decisions are made about which of the multiple criteria the standards rest on.

In selecting criteria on which to base standards for the prevention of diseases caused by more than one factor, the temptation to ascribe an effect of questionable significance and doubtful connection to a pollutant, merely because that pollutant was selected for investigation, must not be yielded to. All too often in the past this commandment has not been followed, particularly in those cases in which a disease entity was not sharply defined or when there was a paucity of data on it.

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